COURSE 2 · SECTION 5 · WORKED SOLUTION

Stack, safety factor and Cpk

Section 5 (videos S05.1 and S05.2) · Tolerance stacks WC vs RSS, FEA safety factor, Cpk

DESIGN QUESTION

Does the shaft end float stack (STK-02) meet its requirement by worst case and by RSS? Can the seat ring OD tolerance alone make STK-01 pass worst case? Which load case and which machine have the least margin?

Steps

1. Open Section06_1_Start.pbix (or Engxtech_S05_2.pbix) > File > Save as > Engxtech_S05_Exercise.
2. Stacks page: SeatRingBuild slicer (single select) A/B, then C. The stack table shows StackID, Requirement, Req Min/Max, WC Min/Max/Result, RSS Min/Max/Result.
3. OD Tol slider (what-if parameter, 0 to 0.02, step 0.001, default 0.015): move it to 0.000 and read the what-if table for STK-01.
4. FEA & Cpk page: card Cases Below Target; bar chart FEA[Case] by Min SF with the 1.5 line; column chart ManufacturingCpk[MachineID] by Min Cpk with the 1.33 line.
5. Right-click _Measures > New measure: Min Cpk Margin (below). Add it to the tooltips of the Cpk chart.

DAX from the video, every stack measure in full (ready to copy)

```
SeatRingBuild = Enter data table, column Build: A/B, C (no relationship)

Stack Nominal =
VAR b = SELECTEDVALUE ( SeatRingBuild[Build], "A/B" )
RETURN SUMX (
    FILTER ( StackContributors, ISBLANK ( StackContributors[Rev] ) || StackContributors[Rev] = b ),
    StackContributors[Direction] * StackContributors[Nominal_mm] )

WC Tol =
VAR b = SELECTEDVALUE ( SeatRingBuild[Build], "A/B" )
RETURN SUMX (
    FILTER ( StackContributors, ISBLANK ( StackContributors[Rev] ) || StackContributors[Rev] = b ),
    StackContributors[Tol_mm] )

RSS Tol =
VAR b = SELECTEDVALUE ( SeatRingBuild[Build], "A/B" )
RETURN SQRT ( SUMX (
    FILTER ( StackContributors, ISBLANK ( StackContributors[Rev] ) || StackContributors[Rev] = b ),
    StackContributors[Tol_mm] ^ 2 ) )

WC Min  = [Stack Nominal] - [WC Tol]       WC Max  = [Stack Nominal] + [WC Tol]
RSS Min = [Stack Nominal] - [RSS Tol]      RSS Max = [Stack Nominal] + [RSS Tol]
Req Min = MIN ( Stacks[MinGap_mm] )        Req Max = MAX ( Stacks[MaxGap_mm] )
WC Result  = IF ( [WC Min] >= [Req Min] && [WC Max] <= [Req Max], "PASS", "FAIL" )
RSS Result = IF ( [RSS Min] >= [Req Min] && [RSS Max] <= [Req Max], "PASS", "FAIL" )
WC Colour  = IF ( [WC Result] = "PASS", "#4D7F3A", "#B5400F" )
RSS Colour = IF ( [RSS Result] = "PASS", "#4D7F3A", "#B5400F" )

WC Tol (what-if) =
VAR b = SELECTEDVALUE ( SeatRingBuild[Build], "A/B" )
RETURN SUMX (
    FILTER ( StackContributors, ISBLANK ( StackContributors[Rev] ) || StackContributors[Rev] = b ),
    IF ( StackContributors[Feature] = "Seat ring outer diameter", [OD Tol Value], StackContributors[Tol_mm] ) )
WC Max (what-if) = [Stack Nominal] + [WC Tol (what-if)]
```

DAX FEA and Cpk (video) and the new measure

```
FEA[SF] = DIVIDE ( FEA[Allowable_MPa], FEA[MaxStress_MPa] )
Min SF = MIN ( FEA[SF] )        SF Target = 1.5
Cases Below Target = COUNTROWS ( FILTER ( FEA, FEA[SF] < [SF Target] ) )
Min Cpk = MIN ( ManufacturingCpk[Cpk] )
Machines Below 1.33 = COUNTROWS ( FILTER ( ManufacturingCpk, ManufacturingCpk[Cpk] < 1.33 ) )

-- new in this exercise
Min Cpk Margin = [Min Cpk] - 1.33
```

STK-02 fails worst case at both ends but passes RSS. RSS assumes centred, independent, normal contributors with +/-tol = +/-3 sigma; with 4 or 5 contributors that is optimistic. On STK-01 the other four contributors alone add 0.036 mm, so even

an OD tolerance of 0 leaves WC Max at 0.086 mm on paper, but that upper end is overstated (see below). STK-01 worst case really fails at the LOW end only (-0.001 mm, a possible interference; rev C 0.006, still below 0.010): tightening the OD (ECR-3200) narrows the range but does not lift the low end to 0.010. STK-01 is a diametral clearance, and roundness, runout and position are one-sided terms entered as +/-, so the upper end of the STK-01 range is overstated (the real maximum clearance is lower). The P300 seating load SF rises from 1.53 (rev B) to 1.78 (rev C) because the stress result is lower; the data gives no reason, and it is not an effect of the OD tolerance change. CNC-M03 has the smallest margin that is not negative only if you round; at 3 decimals it is -0.001, below the limit. The closest machine still at or above 1.33 is INJ-02, Cpk 1.412, margin +0.082.

Expected result

Check	Result
STK-02 worst case (mm)	**-0.020..0.420 (fail)**
STK-02 RSS (mm)	**0.0842..0.3158 (pass)**
STK-01 rev C worst case (mm)	**0.006..0.094 (fail)**
STK-01 rev C RSS (mm)	**0.0298..0.0702 (pass)**
STK-01 WC max with OD tol = 0 (mm, req max 0.080)	**0.086**
Lowest SF	**P200-B Drop 1 m 1.40**
Load cases below SF 1.5	**1**
Machines with Cpk below 1.33	**3**
Lowest Cpk	**GRD-02 0.787**
Closest machine still at or above 1.33	**INJ-02, Cpk 1.412, margin +0.082**